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Acute severe combined demyelination

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Abstract. We present a second case in which Guillain-Barré syndrome (GBS) and acute disseminated encephalomyelitis (ADEM) appeared simultaneously, both in acute and fulminant form. The patient, a 10-year-old girl, presented with acute onset of coma and flaccid, areflexic quadriparesis. The elevated CSF protein levels and delayed F waves fulfilled the criteria of GBS and an MRI study revealed extensive multifocal demyelination compatible with a diagnosis of ADEM. Prompt clinical response followed by complete recovery was achieved by treatment with corticosteroids. It is suggested that acute severe combined demyelination might constitute a separate entity in which the demyelinating process, involving simultaneously the central and the peripheral nervous systems, indicates immune response against a component of the myelin of one system carrying cross-antigenicity with the other.

Key words: Guillain-Barré syndrome – Acute disseminated encephalomyelopathy – Combined demyelination

Central nervous system (CNS) involvement in Guillain-Barré syndrome (GBS) has been reported [3–5, 8, 9]. Usually the predominant clinical manifestations have been those of peripheral neuropathy associated with behavioral changes, papilledema, extensor toe sign, and electroencephalographic abnormalities. Blenow et al. [4] introduced the term "encephalo-myelo-radiculo-neuropathy" to describe this combined involvement. In 1985 Amit et al. [1] reported a case of acute severe combined involvement of both peripheral and central myelin. In this case, GBS was simultaneously combined with acute disseminated encephalomyelopathy (ADEM). We here present a second case of a combined central and peripheral demyelinating process and discuss the possibility that this may constitute a separate pathogenic entity of acute severe combined demyelination (ASCD).

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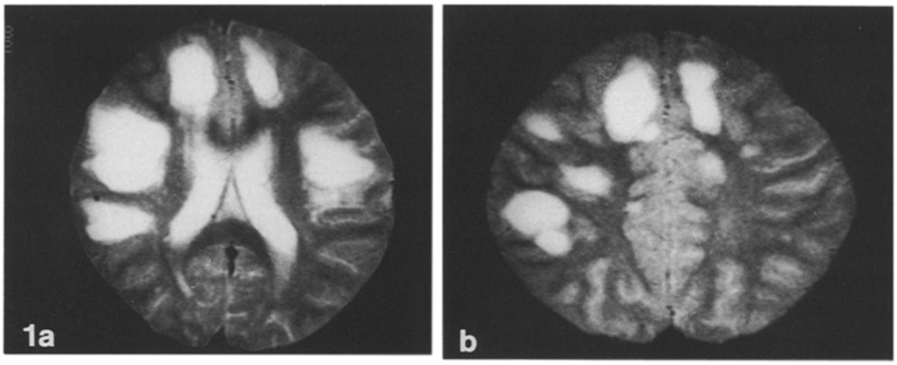
Case report

A 10-year-old girl, generally in good health, was admitted to a local hospital for observation of high temperature, headache, and abdominal and leg pain. A laparotomy performed because of acute abdomen was unrevealing. Recovery from anesthesia was followed by a comatose state that lasted 24 h and flaccid quadriparesis. Two weeks later the girl was transferred to another hospital where the following were recorded: intact mental status, flaccid quadriparesis and areflexia, and lack of sphincter control. Sensory level could not be determined. Examination of the spinal fluid revealed 20 lymphocytes and 50 red blood cells per ml. The protein level was 88 mg/100 ml. Computed tomography (CT) of the head was normal. Urea, glucose, electrolyte, and liver enzyme concentrations were all within normal limits. Testing of urine for porphyrine and heavy metals was negative. Blood cultures and serological tests for polio viruses were negative. Electrophysiological studies of peripheral nerves gave results all within normal limits, except for an absence of F waves. Somatosensory evoked responses revealed low amplitude at the level of the lower brainstem. CT myelography was normal and the spinal repeat tap was acellular with a protein level of 100 mg/100 ml.

With a presumptive diagnosis of a GBS the patient was transferred to a pediatric rehabilitation hospital. A repeated electrophysiological study of the peripheral nerves revealed normal conduction time and absence of F waves in the medial and tibial nerves of both sides. Eight weeks from onset, increased tonus, hyperreflexia, clonus, and upgoing toes were found bilaterally in the lower extremities. Dysarthria and incomplete sphincter control were present. Autonomic phenomena such as fragmented flushing of different parts of the patient's body and paroxysmal sweating were observed. Nine weeks from onset, examination of the fundi revealed bilateral swelling of the optic disks, which was interpreted as optic neuritis. Ten weeks from onset, corneal and gag reflex were absent. The following electrophysiological studies were performed: visual evoked potential – no electrical activity could be traced, nerve conduction velocity was normal, and F response appeared normal. Magnetic resonance imaging (MRI) study of the brain revealed extensive multifocal demyelination (Fig. 1).

At that time, 11 weeks from onset, the patient developed bilateral bronchopneumonia. She was treated with broad-spectrum antibiotics and methylprednisolone 300 mg/day. Within five days of onset of steroid therapy, substantial neurological improvement was observed. Gag and corneal reflexes returned and the hypertonus of the muscles decreased. Two months later, the patient had recovered completely and the neurological examination revealed only fine intention tremor in the left hand. Repeat MRI of the brain revealed a substantial improvement of the demyelination process (Fig. 2).



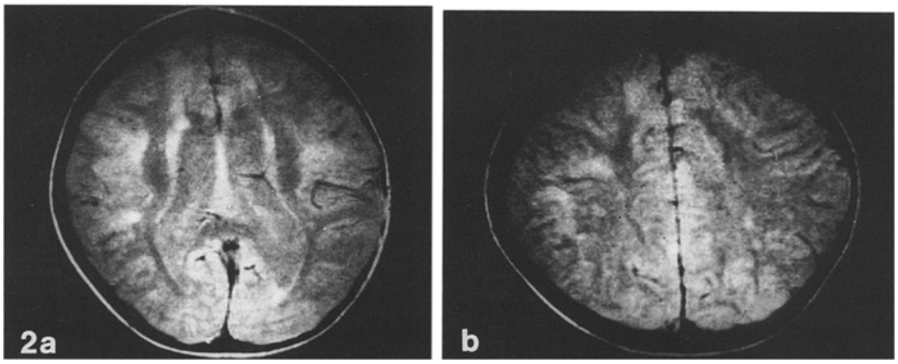


Fig. 1 a, b. Axial MR; T2-weighted images showing extensive multifocal demyelinating areas

Fig. 2 a, b. Axial MR; T2-weighted images 2 months after start of corticosteroid therapy

Discussion

The patient described here presented simultaneous involvement of both central and peripheral nervous systems. Involvement of the CNS was initially manifested by the acute onset of coma and later by the pyramidal signs in the lower extremities. The MRI study supports the diagnosis of ADEM. The involvement of the peripheral nervous system (PNS) was manifested by the flaccid, areflexic weakness lasting for several weeks. The protein-cellular dissociation in the CSF and the absent F waves on electromyelography support the diagnosis of GBS. In the presence of normal nerve conduction velocity it may be argued that myelitis cannot be ruled out. However, the long-lasting areflexic-hypotonic paralysis is highly indicative of peripheral nerve involvement that dominated the clinical picture. During the 11 weeks that elapsed between the onset of symptoms and the initiation of corticosteroid treatment, different stages were noted in the course of the clinical manifestations. It seems that the different stages resulted from the changes in the intensity and extent of involvement of each system (CNS and PNS).

There are only a few disease entities in which substantial involvement of both CNS and PNS exist. Our discussion is limited to those entities of demyelinating origin. Combined myelinopathy is known to occur in dysmyelogenesis of metabolic origin, such as metachromatic leukodystrophy and Krabbe disease. These are hereditary chronic disorders with a well-defined enzymatic defect. Association of chronic inflammatory demyelinating polyradiculopathy was reported in cases of multiple sclerosis [7, 10]. It is not clear whether the simultaneous occurrence of combined central and peripheral demyelination in these patients is a coincidence or whether the two disorders are pathologically related. Willis et al. [12] reported CNS involvement in children with recurrent polyneuropathy. The authors point out that in spite of CNS involvement there was no unequivocal evidence of central demyelination. To the best of our knowledge our two patients with ASCD are the only cases of CNS involvement in GBS with radiological evidence of central demyelination.

Whether ASCD represent a clinical variant of encephalo-myelo-radiculo-neuropathy, as described by Blenow et al. [4], or is a separate pathogenic entity will require further investigation. The following specific characterizations of our two cases support the latter: (**a**) The unusual clinical presentation of simultaneous acute and severe involvement of both PNS and CNS, (**b**) the radiological evidence of central demyelination, and (**c**) the prompt clinical response to corticosteroid therapy (the issue of corticosteroids in either GBS or ADEM is not clear and there are no convincing reports about their efficacy).

The recognition of the possibility of overlapping symptomatology of the two systems is of major importance for early diagnosis of ASCD, since flaccid quadriparesis and hyporeflexia are commonly found in the initial phases of acute encephalopathy. We therefore suggest that electrophysiological studies of the PNS (electromyelography and NCV) be performed in cases where acute encephalopathic clinical signs are associated with hypotonia and hyporeflexia that do not resolve within 72 h. It is also worth noting in this regard that CT of the brain may not always be sufficient in confirming central demyelinating processes in ADEM [2, 6]. In such cases, MRI of the brain may be more sensitive in disclosing the pathology and is therefore recommended.

Pathogenetically, both GBS and ADEM appear to be postinfectious, immunologically mediated attacks on a component of the myelin. It is assumed that in cases with combined involvement of both central and peripheral myelin, an immune response occurs against a component of the myelin shared by the two systems. Two such components of the PNS, P1, which is considered to be identical to CNS myelin basic protein, and galactocerebroside [11], both possessing neuritogenic properties, could function as target antigens in this syndrome.

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